AQA A-Level Further Mathematics (7367)
Question Paper

Further Calculus
E5 - Diff Inv Trig Fns
(A-Level)

Name: ________________________

Class: ________________________

Date: ________________________

Time: 25 min.

Marks: 21 marks

Comments:

Q1.

The function f, where $f(x) = \sec x$, has domain $0 \le x < \frac{\pi}{2}$ and has inverse function f^{-1}, where
$f^{-1}(x) = \sec^{-1} x$

(a) Show that

$$\sec^{-1} x = \cos^{-1} \frac{1}{x}$$

(2)

(b) Hence show that

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{\sqrt{x^4 - x^2}}$$

(4)
(Total 6 marks)

Q2.

(a) Given that $u = \sqrt{1-x^2}$, find $\frac{du}{dx}$.

(2)

(b) Use integration by parts to show that

$$\int_0^{\frac{\sqrt{3}}{2}} \sin^{-1} x \, dx = a\sqrt{3}\pi + b$$

where a and b are rational numbers.

(6)
(Total 8 marks)

Q3.

(a) Differentiate $x \tan^{-1} x$ with respect to x.

(2)

(b) Show that

$$\int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} - \ln\sqrt{2}$$

(5)
(Total 7 marks)